

Chromosome 21 Variant Analysis

Educational NGS portfolio | Sample ERR297183 | Evidence-based technical report

Scope: chromosome 21-only analysis, with targeted review of 20 selected SNVs. Research and training use only; not a clinical diagnostic report.

1. Executive summary

This project documents a chromosome 21-focused NGS workflow and a targeted evidence review. The integrated workbook retains 638 prioritized variant records and a separate 20-variant evidence table. All 20 selected records are VCF FILTER=PASS, and independent pileup-derived allele fractions were compared with VCF DP4-based allele fractions. The analysis is a technical screening exercise, not orthogonal confirmation or pathogenicity classification.

2. Data and analysis workflow

Input: ERR297183 sequencing data and chromosome 21-focused reference/alignment outputs. Available workflow artifacts include a sorted BAM and BAI, VCF files, a prioritized workbook, and a 20-position BED file. The BAM was checked with samtools quickcheck and summarized with samtools flagstat. Selected loci were examined with samtools depth and samtools mpileup (-q 20 -Q 20), then compared with VCF genotype, FILTER, QUAL, MQ and DP4 evidence. The pileup command did not supply a FASTA reference; REF/ALT identities were therefore obtained from the variant evidence table.

Important alignment limitation: use of a chromosome-only reference does not establish genome-wide mapping uniqueness and may permit off-target misalignment. Reported QC statistics must not be interpreted as genome-wide mapping metrics.

3. Summary of observed results

Metric	Observed result
Prioritized variant records in workbook	638
Selected loci reviewed	20 distinct positions
VCF FILTER=PASS	20/20
VCF genotype	0/1: 18, 1/1: 2
Pileup depth (minimum–maximum)	46–456
BAM allele fraction (minimum–maximum)	0.174–0.961
Largest absolute BAM–VCF VAF difference	0.0482 (4.82 percentage points)

The initial flagstat output reported 21,545,691 total records and 1,221,496 mapped records (5.67%) against the chromosome-focused reference. Mean mapping quality among primary mapped reads was reported as 28.50. These are context-dependent technical observations, not a sample-level genome-wide mapping performance assessment.

4. Per-variant evidence summary

The table below is an abbreviated index. Complete forward/reverse REF and ALT counts, VCF quality and mapping quality, depth, pileup VAF and screening fields are preserved in the integrated workbook.

Position (chr21)	REF>ALT	GT	Depth	BAM VAF	VCF VAF
9810517	T>C	0/1	65	0.484	0.456
10473009	A>G	0/1	61	0.344	0.344
10499541	G>T	0/1	238	0.731	0.720

Position (chr21)	REF>ALT	GT	Depth	BAM VAF	VCF VAF
10569701	C>T	0/1	219	0.374	0.376
10595981	G>C	0/1	193	0.244	0.240
10613212	C>G	0/1	73	0.192	0.181
13046374	C>G	0/1	456	0.672	0.624
13065174	C>G	0/1	46	0.848	0.844
13113663	A>C	0/1	341	0.246	0.212
13452189	G>A	0/1	49	0.898	0.889
13452190	C>T	0/1	50	0.540	0.543
17500755	T>C	1/1	153	0.961	0.963
18563314	C>T	1/1	58	0.948	0.948
28887925	G>A	0/1	92	0.174	0.156
28997757	A>G	0/1	46	0.630	0.622
29062383	A>T	0/1	51	0.451	0.451
32504016	G>A	0/1	86	0.791	0.786
32578707	G>A	0/1	73	0.616	0.616
39298488	G>C	0/1	67	0.209	0.200
39298543	G>A	0/1	57	0.298	0.298

5. Interpretation and outstanding review

The selected calls show measurable allele support in the pileup and VCF, with an absolute BAM–VCF VAF difference no greater than approximately 4.82 percentage points in this workbook. This concordance alone does not prove that a call is real. Strand imbalance, local alignment, read-position bias, repetitive regions and reference-context effects require locus-level inspection. Screening flags should not be presented as confirmed artifacts or as evidence of clinical pathogenicity.

A previous screening worksheet categorized 15 variants as requiring targeted follow-up and five as having no screening alert under its applied rules. These are workflow triage categories, not validated pass/fail calls. IGV inspection, genome-wide reference remapping when feasible, population/clinical database checks and orthogonal testing (where warranted) remain outstanding.

6. Deliverables and reproducibility

Primary deliverable: ERR297183_chr21_portfolio_integrated_final.xlsx. Relevant sheets include Prioritized_Variants, Variant_Review, Final_20_Variant_Triage, VCF_Allele_Evidence and Scope_and_Limitations. The full raw VCF sheet (10,943 variants, as documented in the workbook) is not included in this compact deliverable. The BAM is not required to be hosted publicly; scripts, command versions, reference provenance and non-sensitive summary outputs should be documented in the repository README.

7. Limitations and use statement

This is an educational, chromosome-restricted analysis. No genome-wide uniqueness assessment, completed IGV review, orthogonal validation or clinical interpretation is claimed. Pileup and VCF counts may differ because of filtering and counting conventions. Genotype labels are caller outputs, not independent confirmations. Do not use this report for patient care or diagnosis. Confirm that all shared files are free of personal identifiers and comply with source-data licensing and consent requirements.

Source and audit trail

Source of tabulated results: ERR297183_chr21_portfolio_integrated_final.xlsx, inspected for this report. The workbook identifies its upstream sources as ERR297183_chr21_portfolio_review_ready.xlsx, variant_evidence.tsv and ERR297183_chr21_variant_final_triage.xlsx. No external clinical evidence review was performed for this report.